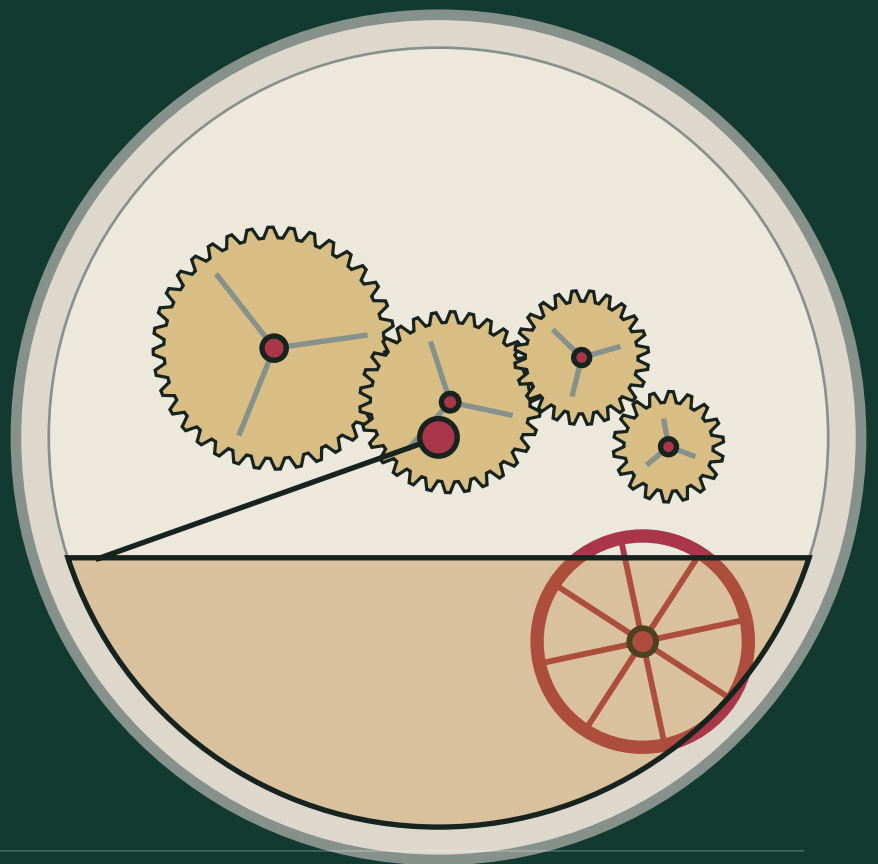


Inside an Automatic Watch

How motion becomes stored energy, measured beats, and displayed time.



ROTOR - MAINSPRING - GEAR TRAIN - ESCAPEMENT - BALANCE

A visual, plain-language introduction to the mechanical movement

START HERE

Four jobs, one continuous chain

An automatic watch is a spring-powered clock that harvests motion from your wrist. Understanding it becomes much easier when the movement is separated into four jobs.

1

CAPTURE + STORE

The rotor turns the winding system. The mainspring stores that work as elastic energy.

2

TRANSMIT

The barrel and compound gear train carry power while creating useful rotational ratios.

3

REGULATE

The escapement meters release. The balance and hairspring establish a repeatable interval.

4

DISPLAY

The motion works convert accumulated gear rotation into seconds, minutes, and hours.

WRIST

SPRING

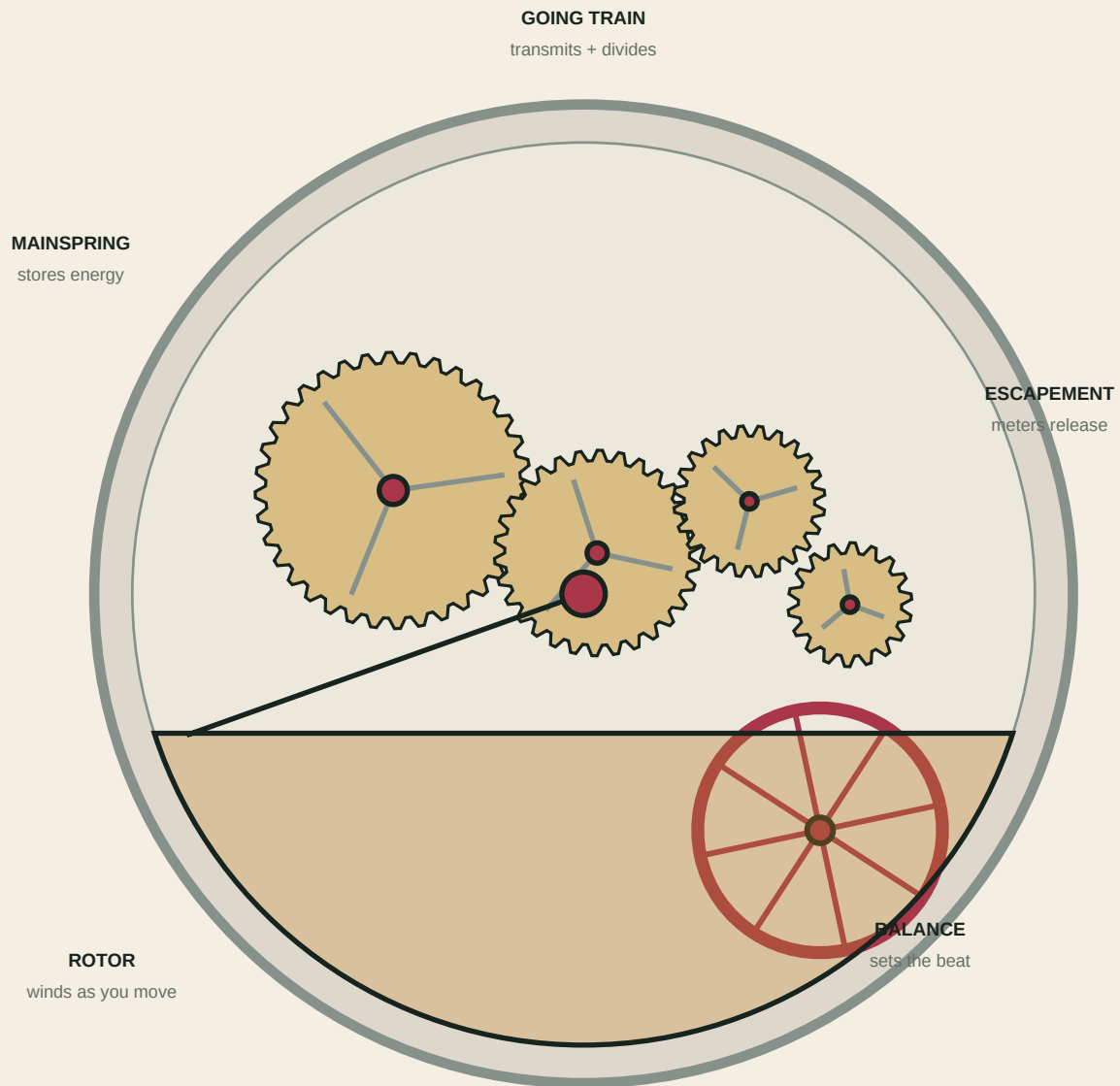
TICK

HANDS

Energy path: wrist movement -> rotor -> winding gears -> mainspring -> going train -> escapement <-> balance -> hands.

CUTAWAY OVERVIEW

The movement as a system



The rotor does not drive the hands directly. It winds the mainspring. The mainspring then drives the watch through the gear train, while the escapement and balance control how quickly that stored energy may leave.

CAPTURE

The weighted rotor responds to changes in wrist orientation and acceleration.

REGULATE

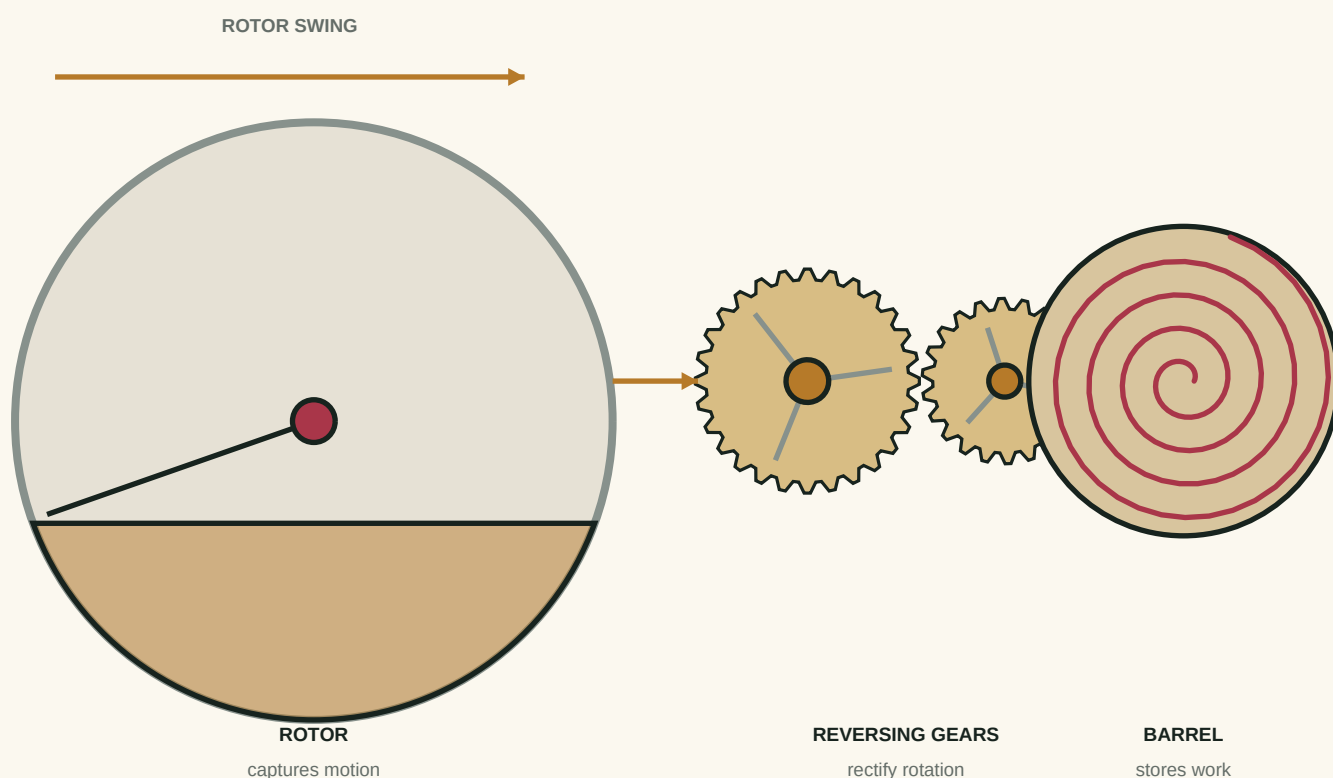
The escapement releases energy in small steps governed by the oscillator.

DISPLAY

The final ratios turn the hands at human-readable rates.

WINDING

From wrist motion to stored energy



What “fully wound” means

The mainspring is a long strip of spring steel coiled inside the barrel. Winding increases its elastic deformation. As it relaxes, the barrel turns and supplies torque to the going train.

AUTOMATIC WINDING

The rotor may wind in one direction or both, depending on the calibre and its reversing system.

OVERWIND PROTECTION

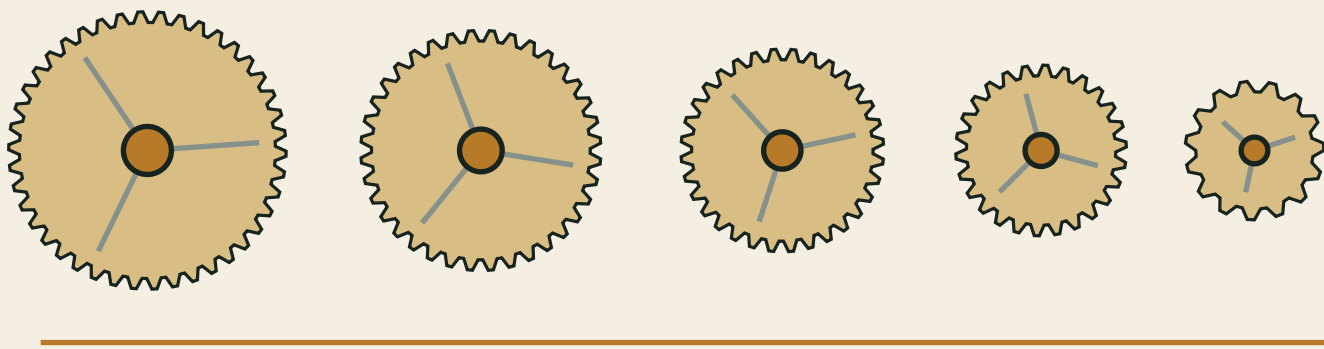
A slipping bridle normally lets the mainspring slide against the barrel wall when fully wound.

POWER RESERVE

A fully wound movement commonly runs for tens of hours; the exact reserve depends on its design.

COMPOUND REDUCTION

The gear train in detail

**BARREL**

slow / high torque

clockwise

CENTER

about 1 rev/hour*

counterclockwise

THIRD

transmission

clockwise

FOURTH

about 1 rev/min*

counterclockwise

ESCAPE

step by step

clockwise

Each large wheel drives a much smaller pinion fixed to the next arbor. This compound arrangement converts the barrel's slow, forceful rotation into the useful rates needed farther through the movement. Every mesh reverses direction.

What the ratios accomplish

SECONDS

The fourth wheel commonly turns once per minute and may carry the seconds hand.

MINUTES

The centre wheel traditionally provides an hourly rotation associated with the minute display.

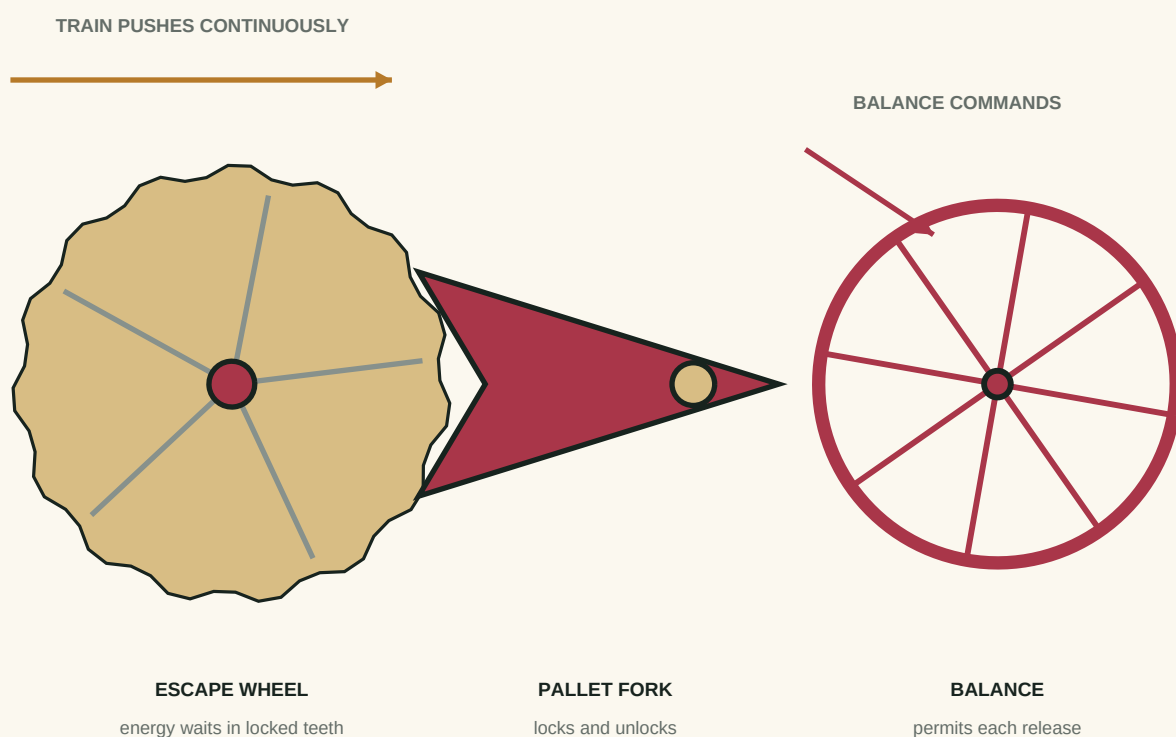
HOURS

The motion works add another 12:1 relationship for the hour hand.

* Conventional arrangement. Exact placement, tooth counts, and seconds architecture vary by calibre.

LOCK - IMPULSE - RELEASE

Why the spring does not unwind at once



1

LOCK

One pallet stone stops an escape-wheel tooth.

2

UNLOCK

The passing balance shifts the fork and frees that tooth.

3

IMPULSE

The tooth pushes the fork, which gives the balance a small sustaining impulse.

4

RELOCK

The opposite pallet catches the next tooth. The train has advanced one step.

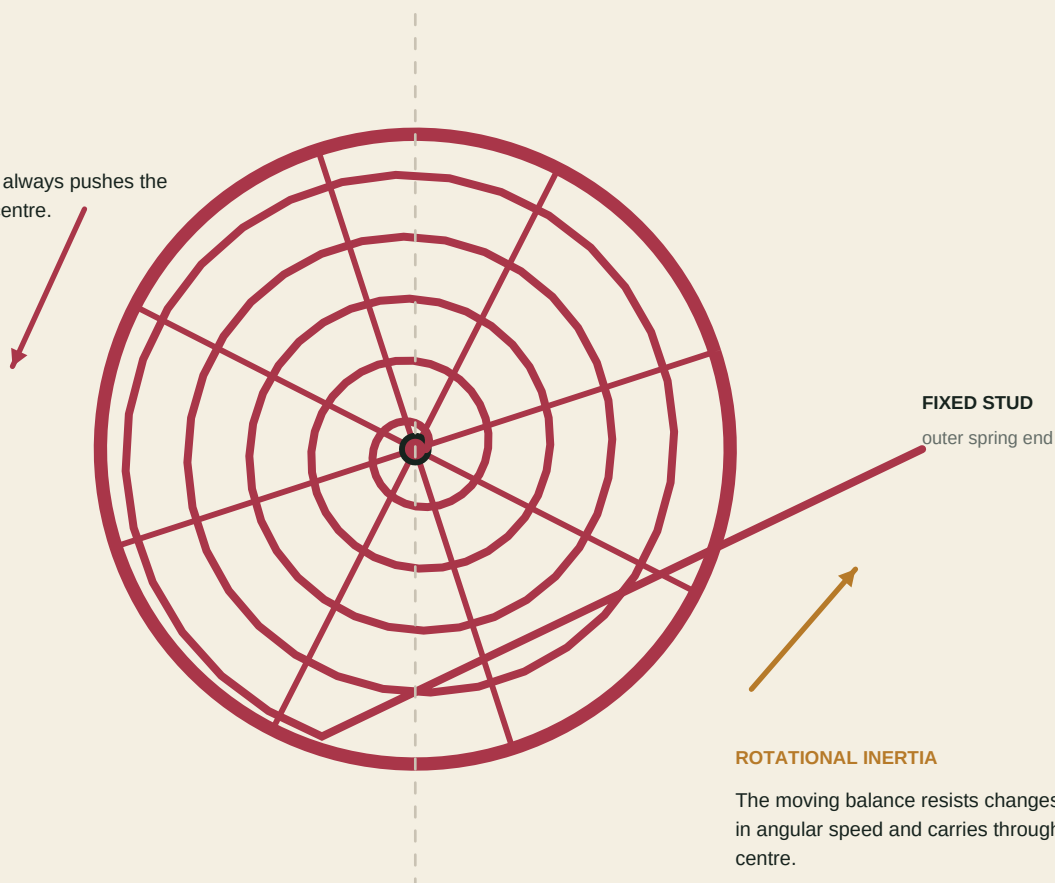
The tick is not the rotor or the mainspring. It is the repeated locking and release of the escapement under control of the oscillator.

THE WATCH'S "PENDULUM"

Balance wheel + hairspring

RESTORING TORQUE

The twisted hairspring always pushes the balance back toward centre.

**FIXED STUD**

outer spring end

ROTATIONAL INERTIA

The moving balance resists changes in angular speed and carries through centre.

A rotational harmonic oscillator

A wristwatch does not use a gravity pendulum. Its time base is a rotating wheel coupled to a spiral spring. The balance supplies rotational inertia; the hairspring supplies a restoring torque. Together they create a natural frequency.

MORE INERTIA

A larger or more outwardly concentrated balance is harder to accelerate, so the oscillator tends to run more slowly.

STIFFER SPRING

Greater hairspring stiffness produces stronger restoring torque, so the oscillator tends to run faster.

AMPLITUDE

A larger arc should not alter an ideal oscillator's period, but real movements show small amplitude-related errors.

ENERGY CHANGES FORM

One complete oscillation

1

CENTRE CROSSING



Spring nearly relaxed

Maximum angular speed



2

OUTWARD



Spring winding

Balance slowing



3

TURNING POINT



Spring maximally twisted

Balance briefly stopped



4

RETURNING



Spring releasing

Balance accelerating

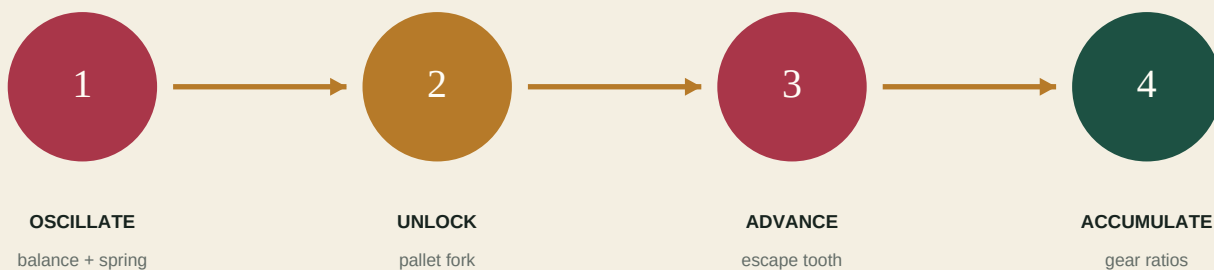


At the centre, most energy is kinetic. Moving outward converts motion into elastic spring energy. At the turning point the balance stops momentarily, and spring energy is greatest. The spring then accelerates the balance back through centre and into the opposite half-cycle.

The escapement adds only a small impulse near centre crossing to replace frictional losses. It sustains the oscillation; it does not create the fundamental interval.

THE COUNTING CHAIN

How equal beats become clock time



28,800 vph

8 vibrations each second

“Vibrations per hour” counts half-oscillations. At 28,800 vph, the escapement performs eight releases per second.

The balance does not know the hour

It produces nearly equal intervals. The escapement turns each permitted interval into a discrete gear advance. The train then accumulates those advances. Fixed ratios make one output turn once per minute, another once per hour, and the hour display once per twelve hours.

ONE DISPLAYED SECOND



Eight small releases - often seen as eight tiny seconds-hand steps at 28,800 vph.

The seconds hand appears to sweep, but it is advancing through discrete steps. Higher beat rates make those steps smaller and more frequent.

WHY WATCHES GAIN OR LOSE

What changes the oscillator's rate

EFFECTIVE SPRING LENGTH

Shortening the active hairspring increases its effective stiffness and speeds the rate; lengthening it slows the rate.

BALANCE INERTIA

Moving regulating mass outward increases inertia and slows the rate; moving it inward speeds the rate.

AMPLITUDE

Low power, friction, positional changes, or shocks can alter amplitude. Real escapements may keep slightly different time at different arcs.

TEMPERATURE + MAGNETISM

Temperature changes spring properties and dimensions. Magnetism can make hairspring coils attract or interfere, producing large errors.

POSITION

Gravity acts differently on pivots and the oscillator in dial-up, dial-down, crown-up, and other orientations.

FRICTION + LUBRICATION

Pivot friction, contaminated oils, and escapement losses change amplitude and disturb consistent energy delivery.

THE REGULATOR'S GOAL

Keep the balance's natural interval and amplitude as stable as possible across time, position, temperature, and mainspring torque.

A FINAL MENTAL MODEL

Follow the energy. Then follow the count.

The movement makes sense when energy flow and time measurement are kept conceptually separate.

ENERGY	wrist -> rotor -> mainspring -> gear train -> escapement -> balance
COUNT	oscillation -> unlock -> tooth advance -> accumulated ratios -> hands

ROTOR

Weighted winding mass moved by the wrist.

MAINSRING

Coiled spring that stores mechanical energy.

BARREL

Drum containing the mainspring; drives the train.

GOING TRAIN

Compound sequence carrying power toward the escapement.

PINION

Small gear fixed to an arbor and driven by a larger wheel.

ESCAPEMENT

Mechanism that locks, unlocks, and impulses.

PALLET FORK

Lever between escape wheel and balance.

BALANCE WHEEL

Rotating inertial element of the oscillator.

HAIRSPRING

Spiral spring producing restoring torque.

AMPLITUDE

Maximum angular excursion of the balance.

BEAT RATE

Number of half-oscillations per unit time.

JEWELS

Hard, low-friction synthetic ruby bearing surfaces.

Core principle: the oscillator creates a repeatable interval; the escapement counts it; the gear train converts the count into displayed time.